

VERIFICATION SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER 35 YEARS OF OPERATIONS

Tukiran Surbakti¹⁾, Endih Pujihastuti¹⁾, Abdul Aziz Rohman Hakim¹⁾, Aslina Ginting²⁾, Muksin Setiawan¹⁾.

1) Research Center for Technology of Nuclear Reactor, Research Organization for Nuclear Energy, National Research and Innovation Agency (BRIN), KST BJ Habiebi, Serpong Gd. 80, Tangerang Selatan, Banten 15314, Indonesia.

2) Research Center for Fuel Fabrication, Research Organization for Nuclear Energy, National Research and Innovation Agency (BRIN), KST BJ Habiebi, Serpong Gd. 80, Tangerang Selatan, Banten 15314, Indonesia.

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*Corresponding author

tukiran@brin.go.id

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Abstract

VERIFICATION OF SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER 35 YEARS OF OPERATIONS. To ensure the safety of RSG-GAS reactor operation, a review of the safety parameters of core operation must be carried out, as documented in the updated Safety Analysis Report (SAR) and Periodic Safety Review (PSR) documents. This research aims to verify the value of safety-related parameters in the SAR (Safety Analysis Report), including neutronic aspects. In this aspect, verification of the reactivity is more balanced than that of the RSG-GAS after 35 years of operation through calculations using the WIMSD-5B / Batan-FUEL codes. From the calculation results obtained, the shutdown margin reactivity is 4.59%, and from experimental results is 4.10%. The deviation of both results is 12% after 35 years of operation. Meanwhile, the excess reactivity parameter on the equilibrium RSG-GAS without xenon of 7.85%. While the excess reactivity value of the RSG-GAS core experiment is 7.11%. From the two values obtained, the calculated value is in accordance with the experimental value, with a difference of 10.4%.

Keywords: Verification, Excess Reactivity, RSG-GAS Reactor, WIMSD-5B, Batan- FUEL

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1.0. INTRODUCTION

Currently, the Multipurpose Reactor (RSG-GAS), the National Research and Innovation Agency (BRIN), as the

Nuclear Installation Operator (NIO) operating the RSG-GAS reactor, will submit a revised Safety Analysis Report (SAR) document to the Nuclear Energy Regulatory Agency (BAPETEN) regarding the reactor's operating permit. In line

with the revision process, BAPETEN, as a technical support unit, is conducting an independent assessment of the RSG-GAS reactor's safety related to its supervisory duties. The results of this study are used to provide technical support to the Licensing Directorate in the SAR evaluation process. This study presents the calculation of the RSG-GAS equilibrium core excess reactivity at the Beginning of Cycle (BOC) condition. This is important to do because the BOC equilibrium core excess reactivity value is a parameter directly related to the core configuration and the control rod's ability to shut down reactor operations [1,2,3]. Therefore, the initial equilibrium core excess reactivity limit is closely related to the reactor's operational safety. The calculation of the BOC equilibrium core excess reactivity was carried out using the WIMSD-5B/Batan-FUEL codes [4,5]. This study aims to obtain the excess reactivity value at the initial equilibrium core without xenon using the WIMSD-5B/Batan-FUEL code using ENDFB VIII.0 nuclear data [6,7,8]. The study was conducted to support the evaluation of the SAR and PSR (Periodic Safety Review) documents for the RSG-GAS reactor, specifically verifying the excess reactivity value at the equilibrium core (BOC). The calculation results will be compared with the experimental results of the RSG-GAS working core. This study was conducted as part of the review of the safety analysis report and periodic safety documents for the RSG-GAS core.

2.0. RSG-GAS CORE DESCRIPTION

The G.A. Siwabessy Multipurpose Reactor (RSG-GAS) is the world's first MTR (Material Testing Reactor)-type research reactor to be operated directly with low-enriched uranium (LEU) fuel elements [9,10]. At the time the RSG-GAS design was carried out, only oxide-type LEU fuel elements (U_3O_8 -Al) were available, as shown in Figure 1. Therefore, RSG-GAS initially used oxide fuel with a Uranium density in the meat of 2.96 g/cm^3 with a ^{235}U enrichment of 19.75% [11]. In order to improve reactor performance, the RSG-GAS core has been converted from uranium oxide fuel to uranium silicide. This was done because the use of silicide fuel with a higher Uranium density can increase the length of the reactor's operating cycle [12,13]. The equilibrium core (TWC) arrangement is a configuration of the equilibrium core (TWC). The active core of the TWC consists of 40 standard fuel elements (FB), 8 control fuel elements, one large central irradiation position (CIP) consisting of 2×2 core grid positions, and 4 irradiation positions inside the reactor core, each taking one core grid position, so that the entire equilibrium core consists of 960 fuel element plates, which means it is identical to 45.7 standard fuel elements [14, 15].

a. Reactor Core

The reactor core consists of grids that form a 10×10 matrix. In the active core, the core grid will form an 8×8 matrix that is used as a place for fuel elements, control elements, reflector elements, and irradiation facilities. At one corner, the core is covered by an L-shaped reflector block made of beryllium, which functions as a reflector or reflector to

reflect neutrons that escape from the core [16,17]. Light water (H_2O) as a coolant flows through the core components to absorb heat and is discharged through a secondary cooling system.

b. Fuel element

The RSG-GAS design uses MTR-type U_3O_8 -Al fuel elements with a uranium (^{235}U) enrichment of 19.75% and a density of 2.96 g/cm^3 . Each fuel element consists of 21 plates measuring 625 mm long and 70.75 mm wide, while the active fuel plate measures 1.30 mm thick, 600 mm long, and 62.75 mm wide. The plates used are made of AlMg. Each fuel element contains an average of 250 grams of uranium. However, currently, the RSG-GAS uses 250 g of silicide fuel with the same geometry and other parameters as the design fuel [8].

Under typical working core conditions (TWC), there are 40 fuel elements and a burn-up distribution (burn fraction) consisting of 8 groups with an average initial burn-up of 23.3% and an average final burn-up of 31.3%. In each cycle, five fuel elements and one control element are replaced, with an average burn-in fraction of approximately 56% [18, 19].

c. Control Elements

The control rods used in the RSG-GAS are made of AgInCd (85%, 10%, and 5%) in a fork-shaped structure and are wrapped in stainless steel [20]. Each control element consists of 15 fuel element plates and two neutron absorber blades. During operation, the RSG-GAS uses eight control rods, consisting of seven shim rods and one regulating rod. These control rods are placed at specific positions within the reactor core. The position of the control rods within the core can be seen in Figure 1.

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3.0. METHODOLOGY

Calculations were performed on a balanced core with a nominal power of 15 MW and free of xenon and samarium. The calculation steps are as follows:

3.1. Cell Calculations

The WIMSD-5B program is only capable of one-dimensional neutron transport calculations; therefore, modeling of the core cells is necessary. Cell modeling is used to calculate the generation of group constants. Group constant generation is intended to obtain the average value of the group constants in a cell by homogenizing the cell. Group constant calculations are performed for all core constituent materials under the conditions described above.

a. Fuel Model

The fuel calculation model is a multi-slab model. In this model, one fuel element, consisting of 21 fuel element plates, is made into 21 sequentially arranged slabs of material. The fuel element can be seen in Figure 2. Each slab consists of meat, cladding, and moderator with thicknesses of 0.027 cm, 0.038 cm, and 0.1925 cm, respectively [22, 23]. These slabs have lengths corresponding to the length of the active material. Other materials outside the active length are homogenized and normalized to the active length and are called the "extra region." In the fuel element, the extra region consists of AlMg2 and water, while the "meat" consists of ^{235}U and ^{238}U , the cladding consists of AlMg2, and the moderator consists of H_2O .

b. Control Element Model

The calculation model for the control material is divided into two regions. The first region is the active region, consisting of 15 fuel element plates arranged in 15 sequentially arranged slabs. The control element can be seen in Figure 3. The modeling is the same as for the fuel element, differing only in the extra region. The second region is the absorber region. This region is divided into 9 slabs and 1 extra region. The control rods in the neutron absorber region contain AgInCd and SS-321. To obtain a macroscopic cross-section of this region, it is combined with the first region in the calculations.

c. Non-fissionable Materials

For non-fissionable materials, the model is similar to that for fissionable materials, with a small amount of ^{235}U added to the active slab and then homogenized throughout.

The general steps in reactor calculations are shown in Figure 4. Once the reactor shape is known, the process of preparing cross-sectional data is a crucial step. Macroscopic cross-sectional data can be obtained by generating cross-sectional images of the reactor core material using the WIMSD-5B software package. This program has been widely used for cross-sectional generation, particularly in Europe. This paper requires cross-sectional data for a silicide-fueled RSG-RAS core with a payload of 250 g. The cross-sectional data preparation uses a plate-type fuel modeled with a multi-slab, as shown in Figure 3. The fuel composition is $\text{U}_3\text{Si}_2\text{-Al}$ with an enrichment of 19.75%, calculated based on its elemental composition. Other important data can be seen in Table 1. This multi-slab model and fuel composition are used as input to the WIMSD-5B program. The fuel and coolant regions are homogenized to obtain a homogeneous lattice.

The direct scattering neutron (DSN) method was used to calculate the lattice parameters, dividing the neutron energy into four groups and using 138 mesh points. The cell geometry consists of four materials: fuel, cladding, coolant, and extra regions. The cell geometry was generated as a function of 17 burn-up degrees. The lattice parameters were calculated under the following conditions [27]:

- Hot, Xe, and Sm equilibrium (IFS4).
- Hot, Xe-free, and Sm equilibrium (IFS3).

- Cold, Xe-free, and Sm equilibrium (IFS2).
- Cold, Xe-free, and Sm equilibrium (IFS1).

3.2. Core Calculations

Core calculations were performed to achieve the equilibrium core of the RSG-GAS reactor. Core calculations were performed using the Batan-FUEL program. The flowchart for core calculations using the Batan-FUEL program is shown in Figure 2. The core is modeled in a two-dimensional X-Y geometry. In this calculation, the reactor power was set at 15 MW according to the experiment, and the effect of fuel loading on the RSG-GAS core on excess reactivity was determined. Fuel loading was carried out in stages, leaving 12% of fuel outside the core. Then the reactor was operated, most likely not yet critical but already approaching critical. Then, one by one, fuel was added to the core to obtain data on the amount of fuel in the core at the first critical point. The purpose of obtaining excess reactivity was so that the reactor could operate throughout its entire cycle. Because cycle length is a parameter that affects core reactivity, the core calculation was carried out for a cycle length that met the safety criteria, namely the criterion for the largest control rod stuck in one reactor still being sub-critical. Called a stuck rod, if the control rod with the highest reactivity value does not successfully enter the core, the reactor core must be in a sub-critical condition. If the stuck control rod criterion is not met, then the fuel position is rearranged. Likewise, for the peak power factor, if the maximum PPF value exceeds the specified limit, a reconfiguration is carried out where the position and location of the fuel are rearranged in the core so that the maximum PPF value is reduced. Core calculations were performed for 5 equilibrium tears, namely cores 95, 96, 97, 98, and 99. The RSG-GAS equilibrium core reactivity at BOC is an important condition in reactor core operation. The RSG-GAS equilibrium core reactivity at BOC conditions is a limitation in relation to the analysis of the control rod's ability to shut down the reactor. The determination of the equilibrium core reactivity value was carried out at the beginning of core operation because, conservatively, the equilibrium core reactivity value will ensure the reactor can operate safely throughout one cycle, namely 630 MWD. This core reactivity value will decrease during operation due to the burn-up factor so that the reactivity value decreases more compared to the excess reactivity value at the end of the equilibrium core due to the effect of the reduced amount of uranium in the reactor core.

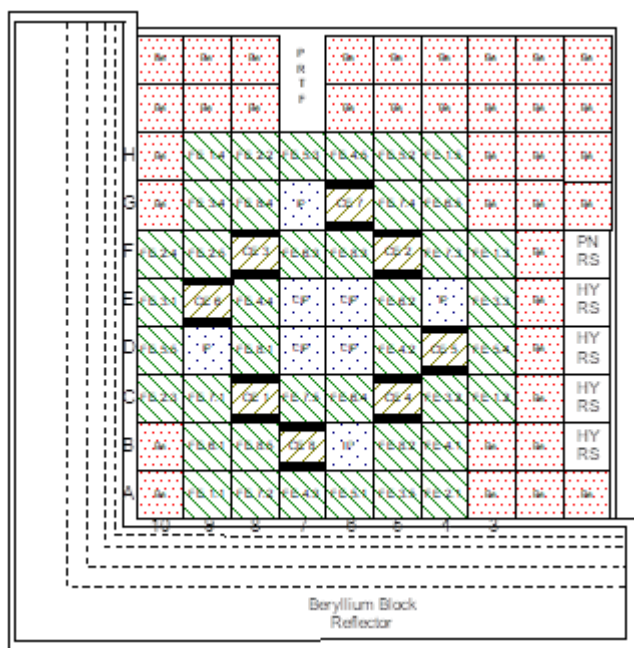


Figure 1 Core configuration of RSG-GAS research reactor [24]

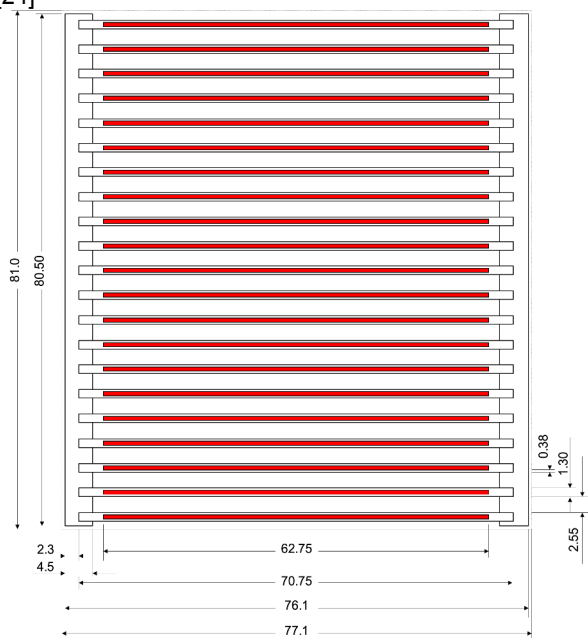


Figure 2 RSG-GAS Standard fuel element [25, 26]

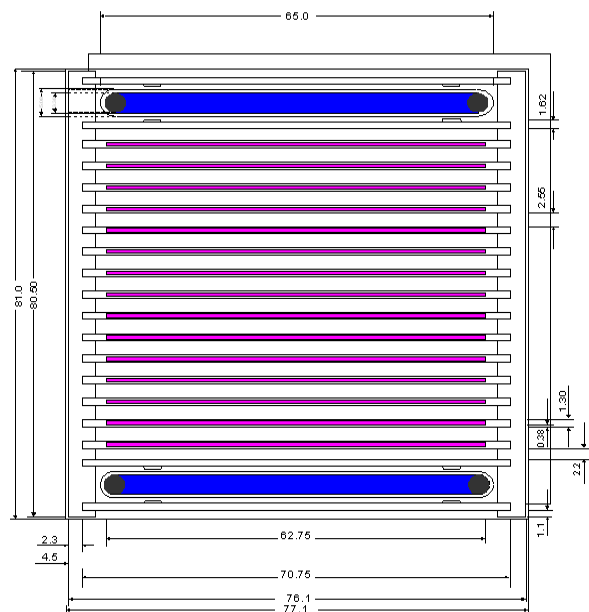


Figure 3 RSG-GAS standard control element [28]

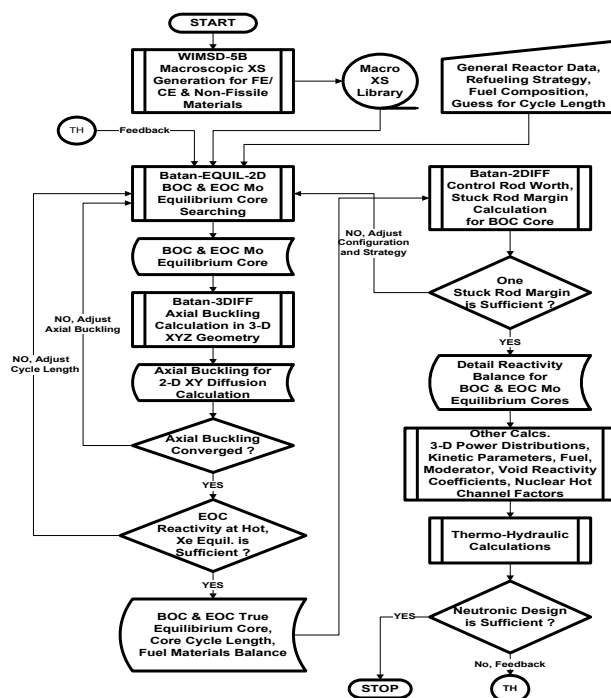


Figure 4 Reactor calculation for RSG-GAS core [29, 30]

4.0. RESULTS AND DISCUSSIONS

Table 1 shows the result of the calculation for the atomic density of uranium silicide. The atomic density of core material is used as data input for WIMS calculation to get the macroscopic cross-section.

Table 1 Atomic densities of the RSG-GAS fuel element.

Component	Nuclide	The density of an atom (Atom/b*cm)
Meat of fuel	²³⁵ U	1.50025x10 ⁻³
	²³⁸ U	6.01895x10 ⁻³
	²⁹ Si	5.01267x10 ⁻³
	²⁷ Al	4.30311x10 ⁻²
Cladding	²⁴ Mg	1.36127x10 ⁻³
	²⁹ Si	1.72395x10 ⁻⁴
	⁶³ Cu	1.26989x10 ⁻⁵
	⁵⁵ Mn	8.81320x10 ⁻⁵
	⁵⁶ Fe	1.15597x10 ⁻⁴
	⁵² Cr	9.31187x10 ⁻⁵
	⁴⁸ Ti	3.37079x10 ⁻⁵
	²⁷ Al	5.76030x10 ⁻²
Coolant	¹ H	6.65620x10 ⁻²
	¹⁶ O	3.32810x10 ⁻²
Extra region	²⁴ Mg	5.60411x10 ⁻⁴
	²⁹ Si	3.34402x10 ⁻⁴
	⁶³ Cu	9.36022x10 ⁻⁵
	⁵⁵ Mn	1.21957x10 ⁻⁴
	⁵⁶ Fe	9.56977x10 ⁻⁵
	⁵² Cr	4.12492x10 ⁻⁵
	⁴⁸ Ti	2.35808x10 ⁻⁵
	²⁷ Al	4.01204x10 ⁻²
	¹⁶ O	2.02507x10 ⁻²
	¹ H	1.01253x10 ⁻²

Table 2 shows the values of k_{inf} as a function of burnup for WIMS calculation. The cross-section is dependent on the burnup of fuel. As greater burn-up, the atomic density is lower because the atomic density itself shows the amount of uranium in the fuel.

Table 2 k_{inf} of WIMSD-5B for fuel assembly calculations.

Steps	Atomic density (Atom/b-cm x 10 ⁻³)	Burn-up (%)	k _{inf}
1	1.5003	0.0	1.6027
2	1.4988	0.1	9
3	1.4913	0.6	1.5979
4	1.4240	5.0	7
5	1.3345	11.0	1.5387
6	1.2447	17.0	8
7	1.1543	23.0	1.5061
8	1.0648	29.0	4
9	0.9752	35.0	1.4823
10	0.8848	41.0	8
11	0.7944	47.0	1.4574
12	0.7037	53.0	8
13	0.5983	60.0	1.4303
14	0.4797	68.0	5
15	0.3753	75.0	1.4009
16	0.2702	82.0	9
17	0.0149	90.0	1.3686
			4
			1.3325
			9
			1.2924
			2
			1.2471
			5

1.1862
4
1.1035
2
1.0124
9
0.8948
8
0.7044
9

The cross-section form from the WIMS calculation was used as input for the Batan FUEL calculation. Figure 5 shows the results of the Batan-2DIFF calculation, which shows a k_{eff} value of 1, indicating that the initial critical reactor is at step 3. This means that the RSG-GAS core is critical with 31 fuel units and 8 control rods for the five cores (T-95 to 99). To increase reactivity, allowing the reactor to operate for one cycle of 42 days at 15 MW, fuel management calculations were performed, resulting in the addition of 9 more fuel units, for a total of 40 fuel units and 8 control rods. The calculation shows that the amount of ²³⁵U in the core when fully charged is (250 gU x 40 FE) + (8 CE x 178.6 gU) = 11.4288 kg. However, to achieve the initial core configuration, not all of the fuel is fresh; some has already been burned, so the amount of uranium in the core should be less than the above value. Based on the results of Batan-2DIFF calculations, it can be seen in Table 2. From Table 3, it can also be seen that during one operating cycle of the 95th core of the RSG-GAS reactor, the ²³⁵U burned was 0.737 kg for 42 days with a power of 15 MW. Likewise, the average ²³⁵U burned was almost the same for the 96-99th core.

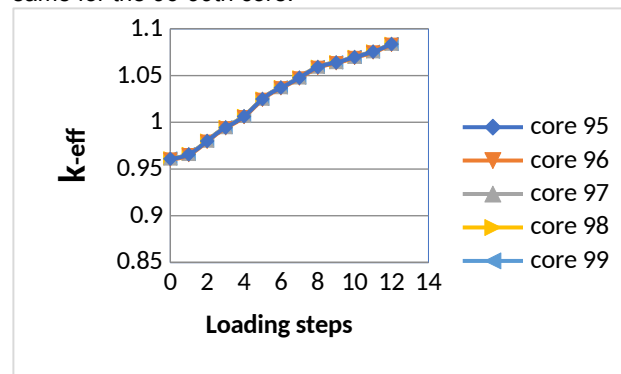


Figure 5: The result of Batan-FUEL calculation

4.1. Calculation of RSG GAS Equilibrium Core Reactivity

Table 3 shows that the RSG-GAS core is initially critical after the introduction of 39 fuels, or in step 3. After obtaining the k_{eff} value from the RSG-GAS initial equilibrium core modeling, the reactivity value must be calculated using the formula below:

$$\rho = (k_{\text{eff}} - 1) / k_{\text{eff}} \dots \dots \dots (1)$$

Based on equation 1 above, by entering the k_{eff} value from the initial equilibrium core modeling of RSG-GAS (T-95) of 1.0847735404, the reactivity value (ρ) is 7.815%, while the experimental result is 7.049%. The deviation between the calculated and experimental results is approximately 10.86%. Table 4 shows the calculated and experimental results for the RSG-GAS cores from the T-95 to T-99

configurations. The table shows that the average deviation for each core is 10.4%. This result is quite good considering the reactor has been operating for over 35 years, and the calculated results have consistently exceeded the experimental results.

The equilibrium core reactivity value (BOC) in the RSG-GAS LAK is 9.2%. This value is based on a design where a single-cycle reactor operates at 750 MWD, meaning the

reactor is operated at 30 MW for 25 days. However, currently the reactor is operated at 15 MW for 42 days (630 MWD), so the higher reactivity must also be adjusted to ensure fuel management continues according to the program. The relative difference is still in 10%, indicating that the verification calculation values between the experiment and the calculation are still consistent.

Table 3: excess reactivity of RSG-GAS core from experiment and calculation.

No	T-95	T-96	T-97	T-98	T-99
0	0.962098002 4	0.962399840 4	0.962576091 3	0.962956786 2	0.963262558 0
1	0.966871380 8	0.967182815 1	0.967338621 6	0.967731237 4	0.968019485 5
2	0.981222271 9	0.981545209 9	0.981605947 0	0.982012987 1	0.982219696 0
3	0.995722115 0	0.996070683 0	0.996080279 4	0.996494948 9	0.996697008 6
4	1.007600069 0	1.007931590 1	1.007968425 8	1.008369088 2	1.008554458 6
5	1.026028156 3	1.026358485 2	1.049177646 6	1.026771903 0	1.026935815 8
6	1.038345456 1	1.038681983 9	1.060580492 0	1.039034366 6	1.039184212 7
7	1.048791647 0	1.049099922 2	1.049177646 6	1.049519896 5	1.049662709 2
8	1.060182094 6	1.060444712 6	1.060580492 0	1.060907602 3	1.061056494 7
9	1.065051794 1	1.065325856 2	1.065442562 1	1.065769553 2	1.065907359 1
10	1.070883631 7	1.071155428 9	1.071266651 2	1.071600794 8	1.071740388 9
11	1.076647162 4	1.076911568 6	1.077029705 0	1.077360034 0	1.077500581 7
12	1.084773540 5	1.085069894 8	1.085151553 2	1.085468292 2	1.085614204 4
Excess reactivity C	7,815%	7,840%	7,847%	7,874%	7,886 %
Excess reactivity E	7,049%	7,218%	7,109%	7,009%	7,168%
D = (C-E) x100 %/E	10,86%	8.61%	10,38%	12.33%	10,01%

*) C= Calculation, E= Experiment, D= Deviasion

Table 4 shows the result of the calculation for the stuck rod condition in the RSG-GAS core after 35 years of operation. The stuck rod condition is still fulfilling the safety operation; that said, the reactor must be sub-critical as one

of the biggest reactivity worth being stuck in the core. From this table, the RSG-Gas reactor is still sub-critical on the stuck rod condition for T95-T99.

Table 4. Stuck-rod conditions of the RSG-GAS core at the Beginning of the Cycle

No.	Condition	T-95	T-96	T-97	T-98	T-99
1	All up -	1.084773540 5	1.085069894 8	1.085151553 2	1.085468292 2	1.085614204 4
2	All dn -	0.955738842 5	0.955906689 2	0.956016719 3	0.956263065 3	0.956397712 2
3	7 down G-6 up	0.978547990 3	0.978663861 8	0.978908717 6	0.979146361 4	0.979354143 1
4	7 down F-8 up	0.982860267 2	0.983050108 0	0.983133971 7	0.983376383 8	0.983489871 0
5	7 down F-5 up	0.977111578 0	0.977198481 6	0.977422773 8	0.977674841 9	0.977899611 0
6	7 down E-9 up	0.981001138 7	0.981252551 1	0.981268525 1	0.981512665 7	0.981592118 7
7	7 down D-4 up	0.979055047 0	0.979204475 9	0.979414343 8	0.979650974 3	0.981592118 7

8	7 down	C-8 up	0.975980579 9	0.976207077 5	0.976201891 9	0.976467967 0	0.976578772 1
9	7 down	C-5 up	0.980520427 2	0.980758786 2	0.980867266 7	0.981151580 8	0.981257915 5
10	7 down	B-7 up	0.978070735 9	0.978329658 5	0.978324532 5	0.978650689 1	0.978782296 2

From Table 4, it can be seen that the shutdown margin value can be calculated based on the position of the control rods, all of which are below, with Equation 1. According to SAR, each reactor core must provide a negative reactivity called a shutdown margin sufficient to shut down the reactor in an emergency. The shutdown margin reactivity value is derived from the results in Table 5. Table 6 shows some

neutronic parameters from experimental results, including the shutdown margin. The results show that the calculation value from the WIMS/Batan FUEL code differs little after more than 35 years of operation compared to the experimental results. This indicates that the RSG-GAS reactor is still feasible to operate, irradiate samples, and produce radioisotopes for research.

Table 5 parameter neutrik teras reaktor RSG-GAS (T95-T99)

Core number	CR pos BOC (cold, clean)	CR pos EOC (hot, poisson)	Energy per cycle (MWD))	Fuel loading (FE/CE)	ρ CR Total (%)	ρ Core excess (%)	ρ Shut down (%)	ρ CR max (%)	ρ Shut down margin (%)
1	2	3	4	5	6	7	8	9	10
95	280/280	548/537	625.64	5/1	-12.92	+ 7.04	-5.88	- 1.86	- 4.02
96	276/276	555/553	625.01	5/1	-13.02	+ 7.21	-5.81	- 1.85	- 3.96
97	275/275	511/511	625.00	5/1	-12.91	+ 7.10	-5.81	- 1.84	- 3.97
98	286/286	541/546	625.01	5/1	-13.20	+ 7.00	-6.20	- 1.85	- 4.35
99	284/284	546/547	625.08	5/1	-13.27	+ 7.16	-6.12	- 1.90	- 4.22

Table 6 shows the average deviation of the shutdown margin of the neutronic parameter in the RGS-GAS reactor between calculation and experimental results after more

than 35 years of operation. The different come from the actual burn up in fuel, they are difficult to estimate when calculating the result.

Table 6 Comparison between calculation and experiment results of shutdown margin for RSG-GSAS cores

Core	Shutdown margin from experiment (%)	Shutdown margin from calculation (%)	Deviation *100%/E)	(C-E)	Remark
T-95	-4.02	-4.631	15.19%		Average of deviation is 12.22%
T-96	-3.96	-4.613	16.49%		
T-97	-3.97	-4.601	15.89%		
T-98	-4.35	-4.574	5.15%		
T-99	-4.22	-4.559	8.38%		

Calculating the thermal neutron flux in the RSG-GAS core irradiation facility is the ultimate goal of reactor utilization. The thermal flux in the D-6, D-7, E-6, and E-7 irradiation areas is crucial because these are the irradiation facilities in the center of the core, where the highest thermal neutron flux is expected. Table 8 shows that the thermal neutron flux in the center of the core irradiation facility is highly dependent on the amount of fuel in the core. The

greater the amount of fuel, the lower the neutron flux. This means that the amount of uranium in the core determines the thermal neutron flux in the irradiation facility. The greater the amount of uranium in the core, the lower the thermal neutron flux. This is inversely proportional to the operating cycle length. The greater the amount of uranium in the core, the longer the cycle length, so the optimal value is used in the design of the research reactor core.

Table 7. Result of calculation for thermal neutron fluxes at the irradiation position ($\times 10^{14}$ n/cm²s)

Steps	E-7	E-6	D-7	D-6	D-9	B-6	E-4	G-7	Loading
0	2.2363	2.2342	2.2196	2.2142	1.6647	1.6081	1.5289	1.9567	0
1	2.1586	2.1552	2.1538	2.1494	1.6074	1.8464	1.4607	1.8105	A-4
2	2.0291	2.0118	2.0399	2.0191	1.6882	1.9979	1.2196	1.5203	A-9
3	1.9541	1.9473	2.0008	1.9894	1.5711	1.9856	1.1579	1.3115	B-5
4	1.8668	1.8678	1.9174	1.9172	1.4016	1.9793	1.2688	1.1805	C-3
5	1.8072	1.8269	1.8722	1.8916	1.2338	2.0297	1.4055	1.0356	C-4
6	1.7463	1.7475	1.8148	1.8155	1.4899	1.9151	1.2394	0.9911	C-10

7	1.7058	1.7214	1.7692	1.7838	13382	1.8566	1.4215	0.9419	D-3
8	1.6502	1.6859	1.6899	1.7231	1.1764	1.6791	1.6246	0.9587	F-3
9	1.6326	1.6517	1.6639	1.6814	1.2831	1.5906	1.5238	1.0241	F-10
10	1.6299	1.6385	1.6424	1.6506	1.2968	1.4875	1.4651	0.9938	G-8
11	1.6022	1.6055	1.6056	1.6088	1.2854	1.3792	1.4024	1.1253	H-8
12	1.5544	1.5541	1.5539	1.5537	1.2854	1.2494	1.3005	1.2404	H-9

The calculation results from the Batan-2DIFF program in Table 7 are in accordance with expectations: in the center area, the average thermal neutron flux is 1.5×10^{14} n/cm²s at a power of 15 MW. In the irradiation position areas D-9, B-6, E4, and G-7, it is even smaller, namely an average of

around 1.2×10^{14} n/cm²s. These calculation results are in accordance with expectations, where IP is used for topaz irradiation while CIP is used for Mo and other radioisotopes irradiation.

A	K	BS+59	B-29	B-30	PRTF	B-20	B-13	B-8	BS+10	B-5	B-2	
	J	B-28	BS+58	B-22	PRTF	B-21	B-23	B-24	B-4	BS+52	B-15	
	H	B-26	1.0161 1.0161 1.0156 1.0137 1.0152	0.9603 0.9588 0.9590 0.9577 0.9573	0.9176 0.9177 0.9210 0.9210 0.9205	0.9305 0.9286 0.9342 0.9328 0.9331	0.8372 0.8366 0.8407 0.8414 0.8412	0.9880 0.9849 0.9885 0.9865 0.9859	B-19	B-17	BS+51	
	G	B-16	1.0571 1.0565 1.0546 1.0543 1.0537	0.9030 0.9014 0.8996 0.9002 0.8995	AL-4	0.9727 0.9733 0.9753 0.9751 0.9776	0.8577 0.8549 0.8575 0.8566 0.8630	0.8782 0.8769 0.8780 0.8772 0.8825	B-40	BS+57	B-14	
	F	1.1164 1.1157 1.1145 1.1136 1.1115	1.1618 1.1619 1.1594 1.1595 1.1576	1.1838 1.1846 1.1834 1.1832 1.1830	1.1200 1.1231 1.1242 1.1274 1.1282	0.9521 0.9524 0.9466 0.9490 0.9508	1.1971 1.1966 1.1991 1.1985 1.1984	0.9211 0.9139 0.9183 0.9187 0.9227	1.0529 1.0509 1.0524 1.0518 1.0504	B-32	PNRA	
	E	1.0536 1.0607 1.0580 1.0549 1.0537	1.0358 1.0369 1.0351 1.0367 1.0361	1.1887 1.1959 1.1931 1.1914 1.1916	AL-6	AL-3	1.1268 1.1247 1.1269 1.1299 1.1322	AL-8	1.0426 1.0411 1.0418 1.0413 1.0408	B-34	HYRA	
	D	0.9962 0.9959 0.9935 0.9979 0.9961	AL-2	1.0373 1.0410 1.0364 1.0358 1.0348	AL-5	AL-7	1.1752 1.1779 1.1795 1.1777 1.1776	1.0878 1.0866 1.0893 1.0891 1.0893	0.8964 0.8948 0.9013 0.8998 0.8987	B-36	HYRA	
	C	1.0278 1.0276 1.0259 1.0241 1.0236	0.9179 0.9179 0.9140 0.9135 0.9107	1.2252 1.2269 1.2254 1.2249 1.2210	1.0035 1.0011 1.0000 0.9987 0.9980	1.0940 1.0960 1.0947 1.1015 1.1000	1.1106 1.1120 1.1127 1.1124 1.1127	1.0924 1.0931 1.0924 1.0920 1.0909	1.0872 1.0864 1.0867 1.0855 1.0837	B-37	HYRA	
	B	BS+54 NS	0.8524 0.8518 0.8506 0.8485 0.8543	0.8110 0.8155 0.8116 0.8119 0.8102	0.9033 0.9044 0.9050 0.9060 0.9058	AL-1	0.9025 0.9038 0.9009 0.8999 0.9007	0.9875 0.9852 0.9921 0.9912 0.9888	B-06	B-11	HYRA	
	A	B-10	0.9495 0.9477 0.9464 0.9471 0.9444	0.7464 0.7438 0.7423 0.7434 0.7457	0.9551 0.9580 0.9576 0.9564 0.9562	0.9676 0.9669 0.9645 0.9711 0.9700	0.9425 0.9466 0.9462 0.9454 0.9457	0.9642 0.9631 0.9614 0.9611 0.9615	B-03	BS+56	B-1	
			10	9	8	7	6	5	4	3	2	1

Figure 6 Calculated radial PPF values for T95-T99 at BOC from Batan-FUEL code

The power peaking factor values are also an important parameter for core analysis. Figure 6 shows the radial PPF values for T95 - T99 cores at BOC. The PPF values are consistent across all cores; the highest value is in the C-8 position, but the highest value differs in every core. For core T95, T-96, T97, T98, and T99 are 1.2250, 1.2269, 1.2249, 1.2210, respectively. They are still below the value that is allowed for the safety limit. The RSG-GAS core is still consistent with the safety limit operation even though the reactor has been operated for more than 35 years. The radial ppf values at BOC for T-95 and T-99 are lower than the maximum limit, namely 1.4 (SAR)

4.0. CONCLUSION

The verification calculations using the WIMSD-5B/Batan-FUEL computer program revealed that the BOC equilibrium overcore reactivity value was in line with expectations, with a 10% difference between the calculated and experimental values. This result is satisfactory and can be used for periodic safety parameter reviews. The calculated RSG-GAS core neutronic parameters are appropriate, and none violate the reactor's operational safety limits. The safety limits include the stuck rod condition, shut-down margin, and overcore reactivity, which must be within the control rod reactivity values.

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